

Use the cross product to find the components of the unit vector  $\hat{n}$  perpendicular to the shaded plane in Fig.

**Solution:**

Two of the plane vectors are  $\mathbf{v}_1 = [-1, 2, 0]$  and  $\mathbf{v}_2 = [-1, 0, 3]$ .  $\mathbf{v}_1 \times \mathbf{v}_2 = \mathbf{n} = [6, 3, 2]$  and  $|\mathbf{n}| = 7$  gives  $\hat{\mathbf{n}} = \frac{1}{7}[6, 3, 2] \square$

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